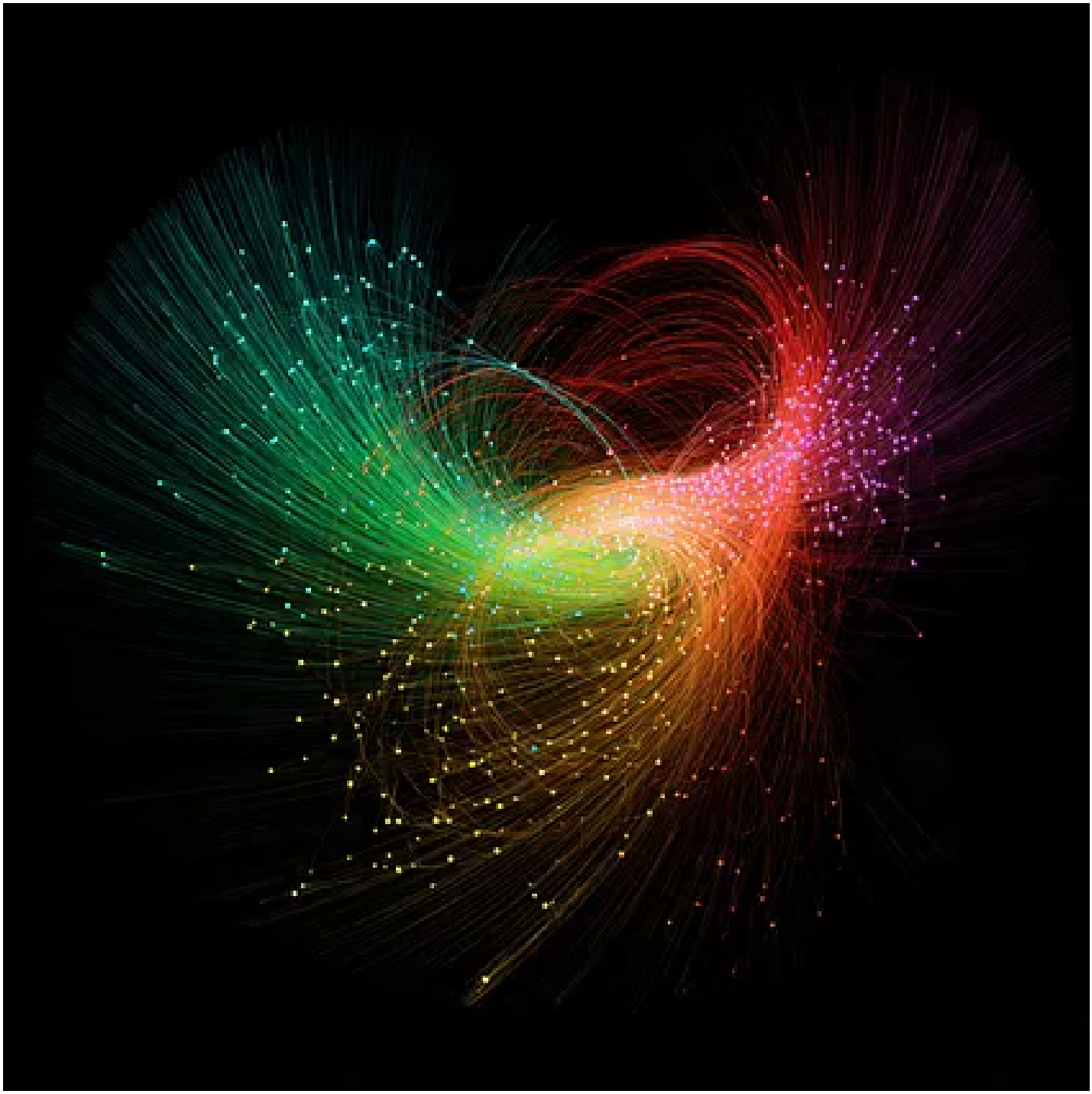
# Introducing the Takens-Based Transformer

An AI Architecture That Treats Language as Trajectories, Not Static Points

KEVIN R. HAYLETT

DEC 17, 2025



*Language as dynamic trajectories rather than static points—the core insight behind the Takens-Based Transformer*

## On the computational cost we've come to accept

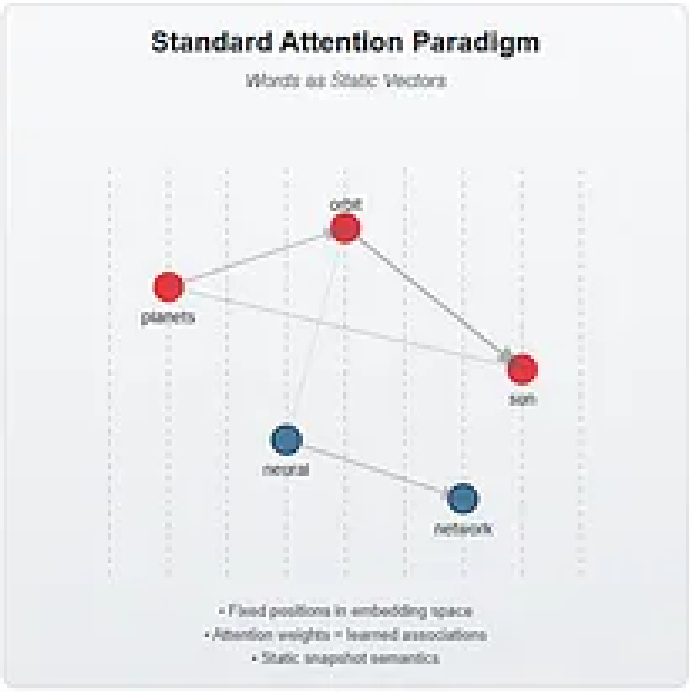
Are you ready? If you've worked with modern language models, you know the technical reality: attention mechanisms that scale quadratically with sequence length, GPU clusters that cost more than a car, and memory requirements that grow relentlessly with every token generated. For a 2048-token context, a standard transformer performs over 4 million comparisons. During generation, it accumulates key-value pairs linearly, creating an ever-growing memory footprint.

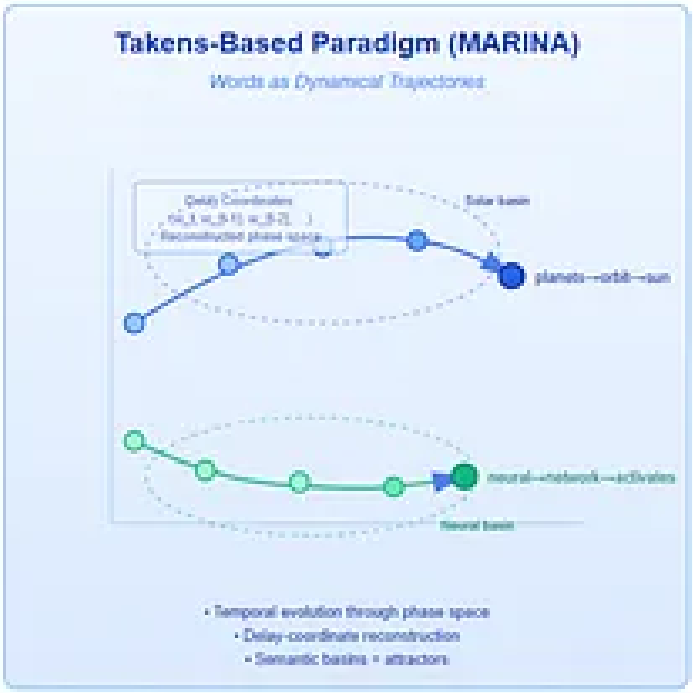
It seems to me we've accepted this computational burden as the necessary price for intelligence. But what my work has shown is there may be a fundamentally different way to model language. This is a method based on 40-year-old mathematics from dynamical systems theory. This theory enables linear complexity, fixed memory usage, and can run on a standard CPU.

> To move beyond theory and show how this could work in practice, I built a fully functional proof-of-concept language model and it works, surprisingly well.

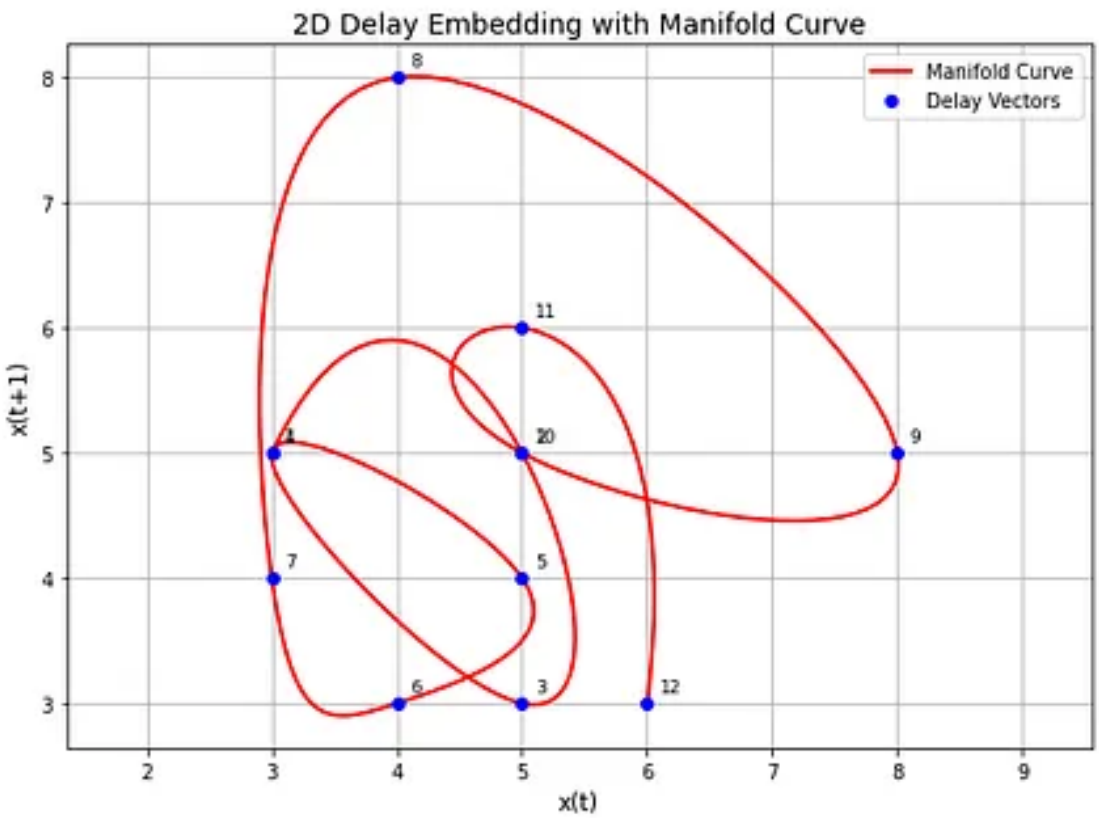
## Language Isn't Static—So Why Model It That Way?

The dominant paradigm in language modelling treats meaning as a property of **static semantic vectors**. Words are points in space; sentences are collections of points; attention mechanisms retrieve relevant context by comparing every point to every other point. If you work in ML you'll have seen the pictures - vectors point into a 'latent' space, showing how words sit next to each other.




But for me language isn't static; it's **dynamic**. So, when you read "The quick brown fox jumps over the lazy dog," you don't process it as seven isolated points. You read it dynamically as a *trajectory* i.e. a path through semantic space that starts at "The," gains momentum through "quick," curves through "brown," arrives at "fox," and so on.




As we can see above, a sentence is no longer a list of words; it's a path through what is called phase space - and in this case follows "The quick brown fox jumps over the lazy dog happily

today before tea". The meaning is not contained in the individual tokens, but in the overall shape of the trajectory's journey. This journey is captured by the neural network in the Takens-based Transformer.

From a dynamical systems perspective, tokens aren't locations. They're **control parameters** that steer a trajectory through semantic phase space. Context isn't something you look up; it's encoded in the *current position and momentum* of that trajectory.

> This shift in perspective, from static points to dynamic trajectories, isn't just philosophical. It's mathematically precise, and leads to a meaningfully different architecture.

## Enter Takens' Theorem: The 1981 Breakthrough That Could Transform AI

In 1981, mathematician Floris Takens proved something very interesting: you can reconstruct the complete state of a dynamical system by observing just **one measurement** over time.

Imagine you only have temperature readings from a single weather station. Takens showed that by creating "delay coordinates" i.e. by concatenating current measurements with measurements from specific past times, you can reconstruct the entire weather system's hidden structure. The technical term is that the delay embedding becomes **diffeomorphic** to the original system's manifold. Here's the formula:

Where h(x(t)) is your single observable measurement (like temperature), and $\tau\tau$ is a delay interval. For 40 years, this theorem has powered chaos theory, time series prediction, and signal processing. But until now, no one has systematically applied it to language modelling.

## The Insight: Attention as Implicit Delay Embedding

In my earlier theoretical work (*Pairwise Phase Space Embedding in Transformer Architectures*), I argued that attention mechanisms don't actually perform "attention" in the human sense. Instead, they're implicitly approximating **delay-coordinate reconstruction**.

Think about it: when a transformer compares a query to all previous keys, what it's really doing is sampling the sequence's history at multiple time points. The attention matrix isn't a relevance scoreboard, from this perspective it's an **approximate Takens embedding**.

> This realization led to a bold question: **What if we replaced attention entirely with explicit delay embedding?**

## MARINA: The Manifold-Aware Reconstruction and Inference Network Architecture

*The Takens-Based Transformer replaces quadratic-complexity attention with linear-complexity delay embeddings*

I implemented this idea in a system called MARINA (Manifold-Aware Reconstruction and Inference Network Architecture). Here's how it works:

### 1. Exponential Delay Embedding

Instead of comparing tokens to all previous tokens, MARINA constructs delay coordinates using **exponentially spaced samples**:

This isn't arbitrary. Language operates at multiple temporal scales:

- **Recent tokens (1, 2)**: Capture syntax and immediate dependencies
- **Medium-range (4, 8, 16)**: Capture sentence and clause structure
- **Long-range (32, 64, 128)**: Capture narrative and thematic coherence

Exponential spacing allocates representation capacity efficiently across all these scales.

### 2. Fixed Memory, Linear Complexity

The implementation uses a **circular buffer** of size 2k+1 where k is the maximum delay power. For delays up to 32, we need just 64 embedding slots. Memory usage is **O(1)**—it doesn't grow with sequence length. Computational complexity is **O(log N)** per token.

Compare this to transformers, where memory grows **O(N)** and computation grows **O(N²)**.

### 3. Adaptive Manifold Projection

The high-dimensional delay vector gets projected onto a learned semantic manifold:

Where *WP* is the Adaptive Takens Layer. This matrix learns which temporal scales matter for prediction—a form of **interpretability** unavailable in attention-based models.

### 4. Channel Theory: Topological Separation

One of MARINA's most innovative features is **Channel Theory**, which creates topologically distinct regions in the semantic manifold:

- **User Channel**: Where input tokens live
- **System Channel**: Where generated output lives
- **Bridge Channel**: Where internal reasoning lives

## Major implications:

This has several important implications. It provides mathematical guarantees of role separation; this is something reinforcement learning can't offer. It also opens up significant safety possibilities, since language manifolds can be cleanly separated. This could prevent prompt space from dissolving into answer space and ensure no more prompt injection type hacks. Facts could be given their own explicit space, helping reduce 'hallucination.' In short, a new world of possibilities around explicit categorization and controllable semantics may become possible

# Three Experiments and Three Insights

Theory is one thing; implementation is another. But the real test comes when you try to make it work. To explore how MARINA behaves in practice, I ran three small-scale experiments, each designed to probe a different aspect of its capabilities.

### Experiment 1: Brown Corpus—General Language Capability

**Goal**: In this experiment the goal was to find out if explicit phase-space reconstruction learn can general linguistic patterns? Could the model actually learn the structure of language? Would the model converge during training or would the results show that the TBT was just another idea doomed from the outset - and a theory that didn't make it off the ground.

**Setup**:

- 15 million parameters (tiny by modern standards)
- Trained on CPU (Intel i7, 32GB RAM)
- 55,542-word vocabulary from the Brown Corpus

**Results**:

- Stable convergence over 44 epochs
- Final validation loss: 4.21 (perplexity ~67)
- Generated coherent, grammatically correct text
- **No attention mechanisms, no GPU clusters**

So the initial conclusion, the theory passed a major test as the model learned to produce well-formed sentences and maintain local syntactic consistency using **only delay-coordinate reconstruction**. And importantly, potential efficiencies were revealing themselves as:

*Training took just 14-17 minutes per epoch on consumer hardware.*

## Experiment 2: Solar System Q&A—Precision Through "Memory Fibres"

**Goal**: In this experiment the major goal was to explore how does the TBT model handle factual recall versus creative generation? This is a vital area for current research especially as modern LLMs are prone to what we call 'hallucination'. That is filling in unknowns with incorrect information.

**Setup**:

- 1.23M parameters
- ~500 question-answer pairs about the solar system

- Progressive repetition protocol (1× to 4× exposure)

## An Unexpected Finding: Memory Fibres

The early experiments showed that when the model learns precise factual mappings, it forms **narrow tubular attractors** in the semantic manifold. These are what I call "memory fibres." These are tight geometric pathways that connect questions to answers with minimal deviation. Maybe an important finding was that as identical Q&A pairs were repeated:

- **Training loss improved 85%** (1.02 → 0.16)
- **Validation loss improved 78%** (2.97 → 0.65)
- **Train/validation gap collapsed to 0.01**

This is the opposite of overfitting. In statistical learning, duplicated training data shouldn't improve validation performance. But in geometric learning, repeated exposure **strengthens the manifold structure**, making it more robust for all trajectories. The final validation perplexity was **1.1**—near-certain predictions.

### Experiment 3: Corpus Ancora—Geometric Learning with Duplicated Data

**Goal**: In this experiment the major goal was to test the geometric learning hypothesis with stylistically coherent text. How would the model hold up with text that had been specifically structured with a consistent poetic form.

**Setup**:

- 2.7M parameters
- Engineered mythopoetic corpus with recursive motifs
- Two conditions: single-pass vs. doubled-data training

**The Paradox That Confirms the Theory**:

One of the very encouraging responses was that with doubled data (every sequence presented twice) we improved the results:

- **Validation loss improved 84%** (3.99 → 0.65)

- **Train/val gap decreased 87%** (2.31 → 0.30)
- **Final perplexity**: 1.9 vs. 54.0 in single-pass

This was quite surprising as classical statistical learning appears to say this shouldn't happen. Why? Because duplicated data provides no new information about held-out sequences.

> However, if the model is learning **geometric structure** (attractor basins, manifold curvature, flow dynamics), then repeated exposure strengthens that geometry. The manifold becomes more coherent, benefiting *all* trajectories—including validation sequences.

These experiments were small, but they pointed toward something intriguing: that language might be less about statistical coincidence and more about geometric shape. And within a TBT the shape becomes clearer with repetition, not noisier.

# Why This Matters: Four Possible Implications

## Increases the Practicality of Language Model Research

These small-scale experiments point toward larger implications, not just for model design, but possibly for the field as a whole. Importantly, I ran these experiments on my old PC, and so MARINA runs on **consumer CPUs**. Basic experiments didn't require GPU clusters, or cloud level budgets. The Brown Corpus model trained in hours on an Intel i7. This significantly lowers the barrier to language modelling research by orders of magnitude and made my own experiment very practical.

## Interpretability Through Geometry

Classically, attention-derived weights in large language models are famously opaque. With billions of parameters encoded as logits in high-dimensional space, tracing how responses emerge is extremely challenging. However, manifold geometry is mathematically rich and seemingly more interpretable as we can analyse:

- Attractor basins (for generalization)
- Memory fibres (for precise recall)
- Channel separation (for role isolation)
- Curvature and flow (for semantic dynamics)

### Fixed Memory Enables Unlimited Context

Because MARINA uses a fixed-size circular buffer, its memory footprint **doesn't grow** with sequence length. You could process documents of different lengths without changing the architecture or memory requirements.

### A New Theoretical Lens for Understanding Transformers

The initial success of MARINA suggests that transformers might work not because of attention per se, but because they're implicitly performing **state reconstruction**. Attention could be just one way to approximate what Takens embedding does explicitly. This perspective reframes language modelling as **dynamical system reconstruction** rather than statistical pattern matching.

## The Trade-offs: Precision vs. Generalization

As in many real-world applications, practicalities matter. The memory fibre phenomenon reveals an important trade-off:

- **Narrow tubes**: Enable precise recall but limit generalization
- **Broad basins**: Enable creative variation but reduce precision

This framing may offer some advantages. The architecture learns the geometric structure inherent to the task. Factual Q&A produces tubes; creative writing produces basins. For applications, this suggests that for some circumstances such as Medical/legal Q&A we can train with high repetition for precise memory. Then for usage such as creative writing we could train with variation for broad basins. Then we may also have the option to create mixed domains where we use hybrid strategies during model training and development.

# Limitations and Open Questions

It's clear that this research is currently in its early stages, serving as a foundational proof-of-principle rather than a definitive solution. While the initial results are promising, they highlight many critical questions that need addressing.

Chief among these is the challenge of scalability: specifically, how these mechanisms behave when integrated into billion-parameter models. Then beyond scale, we must investigate the details of implementation, such as determining the optimal delay schedules across diverse domains and exploring the potential of hybrid architectures that combine traditional attention mechanisms with temporal delays.

Finally, there's the intriguing possibility of developing methods to directly visualize the learned manifolds, which would provide much-needed interpretability. Addressing these points will be essential in taking these ideas further and somewhat beyond my current resources. The next logical step for me is to test MARINA at the 100M parameter scale on diverse corpora. However, for that, I'm going to need a bigger boat - err, um, I mean PC.

The complete paper explores these questions in more depth.

# Beyond Proof-of-Principle: Why This Matters for the Real World

However, while the theoretical insights are fascinating, for me, what makes the Takens-Based Transformer genuinely important are its **practical implications. Especially when we consider** energy efficiency, cost, and accessibility.

## Reflecting on AI's hidden energy crisis—and how TBTs might help

The numbers are becoming unsustainable:

- **Training GPT-3**: ~1,287 MWh (power for 120 US homes for a year)
- **Projected 2030 AI energy use**: Could reach 85–134 TWh annually
- **A single GPU server**: 300–1000W continuous draw, plus cooling

MARINA offers a path to **order-of-magnitude improvements**:

## Algorithmic Efficiency (3-10x)

- **Standard Transformer**: $O(N^2)$ complexity → 4.2M comparisons at 2048 tokens
- **Takens-Based Transformer**: $O(\log N)$ complexity → ~7 delay samples
- **Reduction**: ~600,000x fewer comparisons at same context length

### Memory Efficiency (100-1000x)

- **Transformer KV cache**: Grows linearly with sequence length (~14GB at 4096 tokens for 7B model)
- **MARINA buffer**: Fixed size (~0.0002GB regardless of length)
- **Memory doesn't grow** → No "context window" limits in practice

### Hardware Efficiency (5-10x)

- **No GPU dependency**: Optimized for CPU execution
- **Lower power draw**: CPUs more efficient per FLOP for this workload
- **Consumer hardware**: $1,000 workstation vs. $500,000 GPU cluster

## Concrete Energy Savings: Real Numbers

From my experiments with the Brown Corpus model (15M parameters):

| Metric | Traditional Approach | MARINA | Improvement |
|---|---|---|---|
| Hardware | GPU server (300W+) | Intel i7 (65W) | ~80% less power |
| Training Energy | ~5-10 kWh/epoch (est.) | ~0.04 kWh/epoch | ~125-250x less |
| Memory/Token | ~3.5KB/token (growing) | ~0.05KB/token (fixed) | 70x less, no growth |
| Access Cost | $10-100/hour (cloud) | $0.10/hour (electricity) | 100-1000x cheaper |

**Extrapolating to scale**: A modest deployment of 1000 inference instances:

- **Traditional**: 1000 × 300W GPUs = 300kW continuous = 2.6 GWh/year
- **MARINA**: 1000 × 50W CPUs = 50kW continuous = 0.44 GWh/year
- **Savings**: 2.16 GWh annually = power for ~200 homes

## Where TBTs Could Have Real Potential If They Prove Useful

My main goal in creating the TBT was simply to see whether the theory from my earlier work held up to scrutiny and to determine if it was provable in a scientific sense. I believe the results show that it does. However, it's impossible for me to replicate a miniscule fraction of the work that has now been carried out using Attention in the field of ML. So all I can do is close this work and place into the wider space of AI research for others to investigate should they find the ideas and work interesting. However it doesn't hurt speculating on how a novel architecture may be useful! While this research is in its infancy, it seems worth speculating on how such an architecture might reshape our world. What if we are glimpsing a future where the barriers to artificial intelligence are lowered, replaced by a landscape that is not only more efficient but also more equitable and sustainable?

By dramatically reducing the entry cost of language modelling, we could make AI and especially LLMs far more accessible. A modest workstation could become a hub for Edge AI, empowering universities in the developing world to preserve local languages or deploy diagnostic tools in remote clinics. This shift could extend into specialized enterprise settings, where "precision instruments" replace general-purpose bots. Imagine systems built on specialized memory fibres, enabling businesses to handle technical documentation, legal lookups, or medical Q&A with 90% less overhead and far greater accuracy.

Ultimately, it's encouraging to imagine this path leading toward a more sustainable and accessible future for AI. In a world striving for carbon neutrality, meaningful gains in energy efficiency could allow organizations to meet ESG goals without sacrificing performance. And perhaps this vision culminates in the classroom, where students might train models on personal laptops, and researchers can prototype without cloud-scale budgets. In this speculative, yet maybe plausible future, AI becomes an accessible tool for the many, not just the few, opening the door to a new wave of creative and local innovation.

## The "Memory Fibre" Advantage: Specialized Efficiency

This is where TBTs might outperform general LLMs even on accuracy. The early experiments show how the model can build up very well defined memories and we can mark training data as factual and create manifolds of language that are designed for factual recall. Consider that for **factual retrieval tasks** (medical Q&A, technical docs, legal lookup):

- **MARINA**: Forms precise "memory fibres" → near-perfect accuracy after minimal training

- **General LLM**: Requires massive pretraining, still hallucinates
- **Energy difference**: ~100x in favour of TBTs

**Example**: A pharmaceutical company needs a system to answer questions about drug interactions:

- **Option A**: Fine-tune GPT-4 → $100K+, 10kW, occasional errors
- **Option B**: Train MARINA on their documentation → $1K, 0.1kW, deterministic accuracy

## The 10x Challenge: Is It Achievable?

Let's be conservative but also realistic:

**Where TBTs could deliver 10x+ improvements today**:

- **Energy per query** for specialized tasks
- **Hardware costs** for deployment
- **Memory requirements** for long contexts
- **Training costs** for domain-specific models

**Where attention still wins** (for now):

- **General reasoning** across diverse domains
- **Emergent capabilities** at massive scale
- **Few-shot learning** on unseen tasks

Possibly a smart approach may be to use h**ybrid systems** that use each where most appropriate. A corner stone of modern engineering practice is to find the right tool for the job rather than a one-size fits all approach.

## A Practical Path Forward

For me, this isn't about replacing all transformers tomorrow, far from it. Maybe it's more about:

- **Recognizing that efficiency matters**—not just capability
- **Developing specialized tools** for specialized problems
- **Democratizing access** to language technology
- **Reducing environmental impact** of AI progress

MARINA shows that **10x efficiency gains are possible now**—not through incremental optimization, but through fundamentally rethinking assumptions.

## The Long Road Ahead

Of course as I emphasize, this early work is just a proof-of-principle; the experiments reported here are tiny (approximately one to fifteen million parameters) compared to large language models. Therefore, the experiments lie well below the regime of the 'phase transition' to interactive, multi-domain reasoning, and this work at this small scale doesn't demonstrate generalization, cross-domain abstraction, or the emergence of reasoning observed at the ~1.5B parameter scale and beyond. Whether this transition arises from scale, training diversity, or structural features of the attention mechanism itself remains an open question.

It's very possible that attention contributes to this phase change, and combining attention with explicit delay-coordinate reconstruction may be possible. However, the results are a proof of principle and show that a dynamically grounded, non-attention architecture can learn linguistic structure efficiently at small and medium scales, offering a new direction for future researchers!

## Conclusion: A New Path Worth Exploring

For me the Takens-Based Transformer isn't just an alternative architecture. It's a **proof-of-concept** that language modelling can be fundamentally rethought. For 40 years, Takens' Theorem has shown us how to reconstruct dynamics from single observables. For 7 years, transformers have dominated AI with attention mechanisms. MARINA connects these two worlds. The implications may be important:

- **Efficiency**: Linear complexity, fixed memory, CPU compatibility

- **Interpretability**: Manifold geometry over opaque attention weights
- **Theory**: Language as trajectories rather than static points
- **Accessibility**: Research without institutional compute budgets

This work doesn't claim to beat the latest state of the art LLMs. That was never the point. The goal is to show that **other paths do exist. Importantly, these** paths are grounded in rigorous mathematics, and may lead somewhere equally interesting or even more fruitful.

For me, language isn't just patterns to match. It's a dynamical system to reconstruct with trajectories to navigate. And after building a Takens-Based Transformer, I'm more convinced than ever that this dynamical perspective isn't just metaphor as its grounded in mathematics, implementable, and it works.

- *Visit the site web site listed below for more information.*

*Citation*: Haylett, K.R. (2025). "Introducing the Takens-Based Transformer" Substack, Dec 2025.

*Keywords*: Takens Transformer, Nonlinear Dynamics, LLM, Takens Theorem, Philosophy, Geofinitism.

For those interested in developing these ideas, an open-source implementation of the Takens-Based Transformer proof-of-concept will soon be fully available on *GitHub*. The codebase includes:

- Core architecture in PyTorch
- Training scripts for used for the above experiments
- Inference examples and generation scripts

Everything runs on CPU-only hardware. No special dependencies, no expensive compute requirements.

To explore these ideas further:

- Takens-Based Transformer Website
- GitHub Repository of Proof-of-Concept: Open Source
- Paper showing how LLM ‘Attention’ as Takens method of delays
- Language as Trajectories
- The Compression of Language
- JPEG Experiment including the code
- Finite Tractus: The Hidden Geometry of Language and Thought
- Finite Tractus: Principia Geometrica - Coming Soon

> *Omne quod est, finitum est; tantum per mensuram cognosci potest*
> Everything that exists is finite; it can only be known by measure
>
> *kevinhaylett.substack.com* | *geofinitism.com*
> Copyright © 2025 Kevin R. Haylett